

Multiple Choice

1. **Answer:** B

Options: A) Magnetic susceptibility; B) Hall coefficient; C) Electrical resistivity; D) Thermal conductivity

Note: Correct option: B - Hall coefficient

2. **Answer:** B

Options: A) 2; B) 250; C) 5,000; D) 10,000

Note: Correct option: B - 250

3. **Answer:** D

Options: A) As; B) P; C) Sb; D) B

Note: Correct option: D - B

4. **Answer:** D

Options: A) $(1/2)$ k T; B) kT; C) $(3/2)$ k T; D) 3 kT

Note: Correct option: D - 3 kT

5. **Answer:** D

Options: A) 44 Hz; B) 196 Hz; C) 262 Hz; D) 393 Hz

Note: Correct option: D - 393 Hz

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '450 m'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'real'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 5,000 s. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** a dye laser. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.